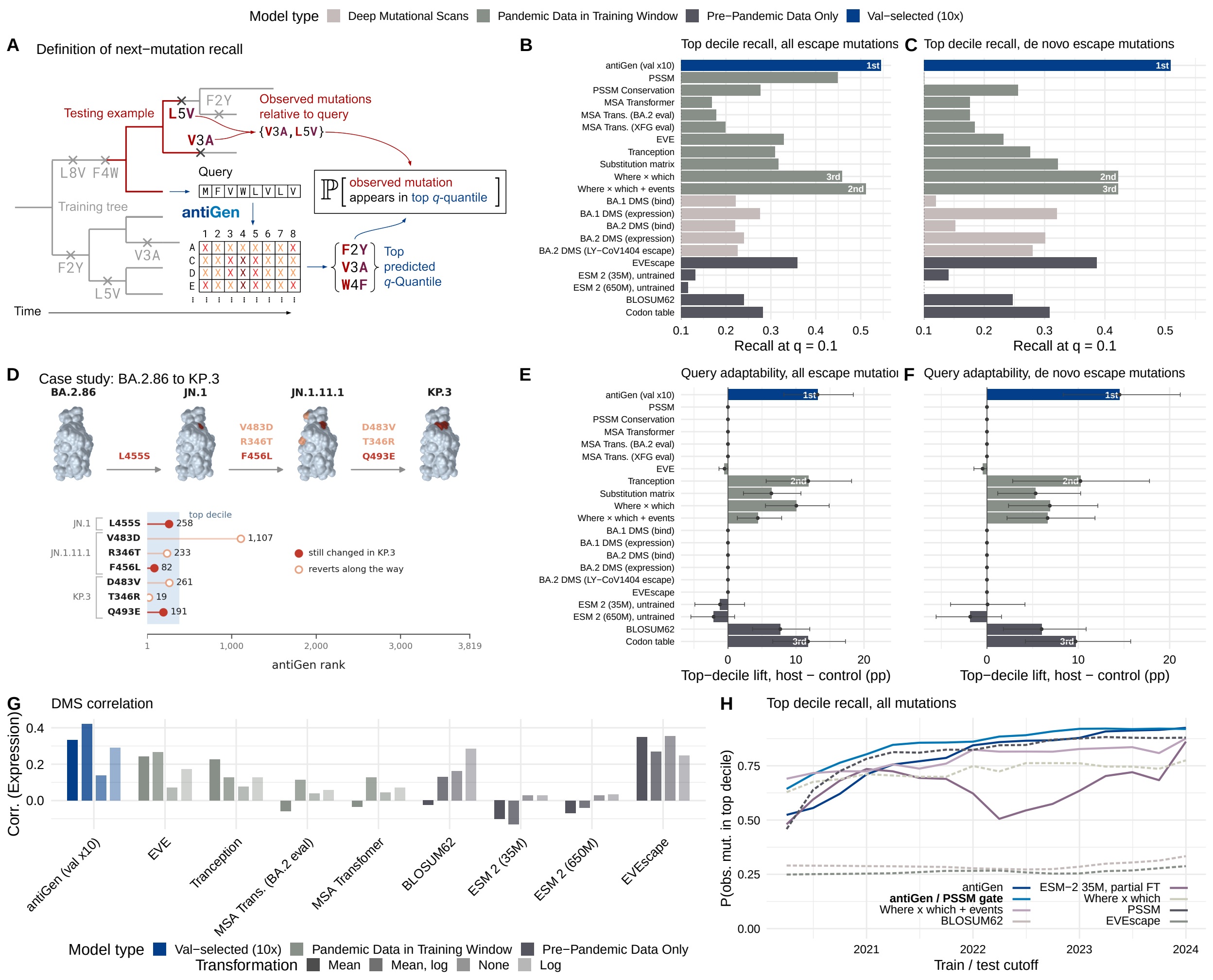

Model type
Deep Mutational Scans
Pandemic Data in Training Window
Pre−Pandemic Data Only
Val−selected (10x)
A
Definition of next−mutation recall
Testing example
L5V
F2Y
V3A
Observed mutations relative to query
{V3A, L5V}
L8V F4W
Query
M F V W L V L V
antiGen
Training tree
V3A
F2Y
L5V
1 2 3 4 5 6 7 8
A
C
D
E
ℙ[observed mutation appears in top q-quantile]
F2Y
V3A
W4F
Top predicted q-Quantile
Time
B
Top decile recall, all escape mutations
antiGen (val x10)
PSSM
PSSM Conservation
MSA Transformer
MSA Trans. (BA.2 eval)
MSA Trans. (XFG eval)
EVE
Tranception
Substitution matrix
Where × which
Where × which + events
BA.1 DMS (bind)
BA.1 DMS (expression)
BA.2 DMS (bind)
BA.2 DMS (expression)
BA.2 DMS (LY−CoV1404 escape)
EVEscape
ESM 2 (35M), untrained
ESM 2 (650M), untrained
BLOSUM62
Codon table
1st
2nd
3rd
0.1 0.2 0.3 0.4 0.5
Recall at q = 0.1
C
Top decile recall, de novo escape mutations
1st
2nd
3rd
0.1 0.2 0.3 0.4 0.5
Recall at q = 0.1
D
Case study: BA.2.86 to KP.3
BA.2.86
JN.1
JN.1.11.1
KP.3
L455S
V483D
R346T
F456L
D483V
T346R
Q493E
top decile
JN.1
L455S 258
V483D 1,107
JN.1.11.1
R346T 233
F456L 82
D483V 261
KP.3
T346R 19
Q493E 191
still changed in KP.3
reverts along the way
1 1,000 2,000 3,000 3,819
antiGen rank
E
Query adaptability, all escape mutatio
antiGen (val x10)
PSSM
PSSM Conservation
MSA Transformer
MSA Trans. (BA.2 eval)
MSA Trans. (XFG eval)
EVE
Tranception
Substitution matrix
Where × which
Where × which + events
BA.1 DMS (bind)
BA.1 DMS (expression)
BA.2 DMS (bind)
BA.2 DMS (expression)
BA.2 DMS (LY−CoV1404 escape)
EVEscape
ESM 2 (35M), untrained
ESM 2 (650M), untrained
BLOSUM62
Codon table
1st
2nd
3rd
0 10 20
Top−decile lift, host − control (pp)
F
Query adaptability, de novo escape mutations
1st
2nd
3rd
0 10 20
Top−decile lift, host − control (pp)
G
DMS correlation
Corr. (Expression)
0.4
0.2
0.0
antiGen (val x10)
EVE
Tranception
MSA Trans. (BA.2 eval)
MSA Transfomer
BLOSUM62
ESM 2 (35M)
ESM 2 (650M)
EVEscape
Model type
Val−selected (10x)
Pandemic Data in Training Window
Pre−Pandemic Data Only
Transformation
Mean
Mean, log
None
Log
H
Top decile recall, all mutations
P(obs. mut. in top decile)
0.75
0.50
0.25
0.00
2021
2022
2023
2024
Train / test cutoff
antiGen
antiGen / PSSM gate
Where x which + events
BLOSUM62
ESM−2 35M, partial FT
Where x which
PSSM
EVEscape